

STUDENT NAME: SOJIB HASAN

FINAL REPORT FOR SEMESTRAL WORK

PROJECT NAME: Personal Calendar - database design

Project Execution

The designed database is a model of a database system for an event management application. The database structure demonstrates operational skills in Microsoft SQL Server and knowledge of T-SQL. The database contains 4 main tables that manage users, event categories, events, and reminders.

Database Creation and Table Description

Users

users - Contains user information including ID, email, name, age, password, occupation, and address

Columns: user_ID, User_Email, User_Name, User_Age, Password, User_Occupation_, User_address, country

Event Categories

categories_of_event - Defines different types of events with their terms and conditions

Columns: ID, name_of_event, Terms_and_codtions, type_of_Event

Events

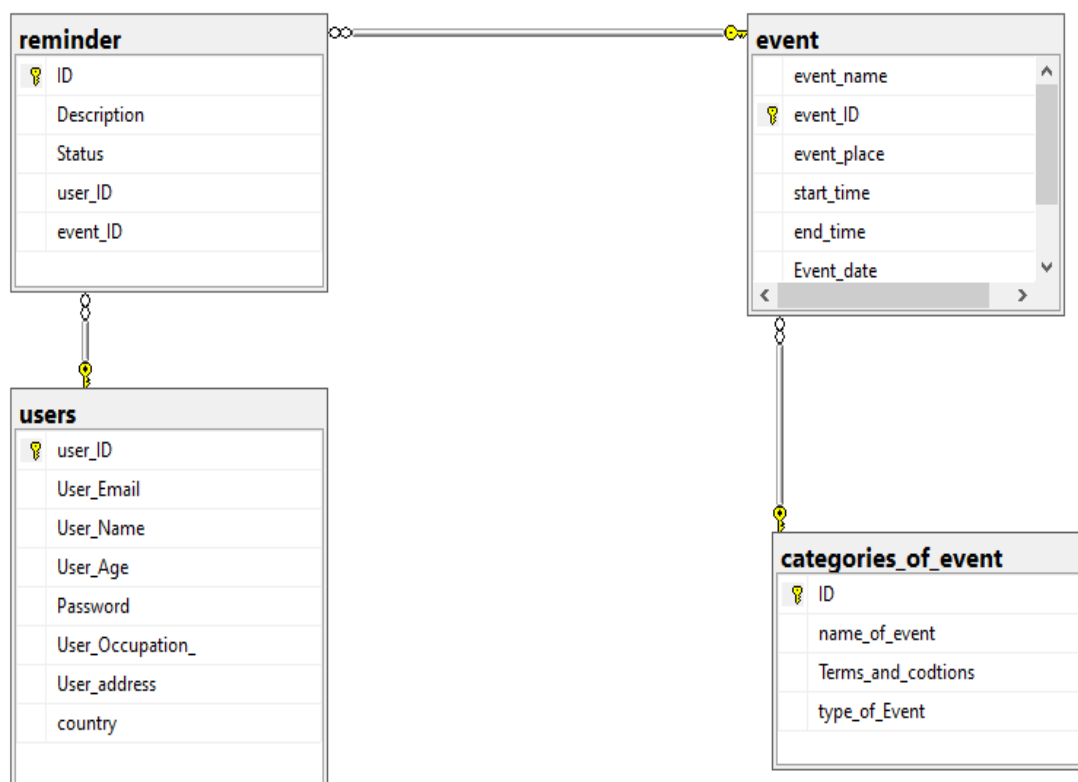
event - Stores details about specific events including timing and location

Columns: event_name, event_ID, event_place, start_time, end_time, Event_date, ID

Reminders

reminder - Tracks reminders sent to users about upcoming events

Columns: ID, Description, Status, user_ID, event_ID



THIS IS THE MAIN DIAGRAM.

Table Creation Code:

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```
CREATE TABLE users
(
    user_ID INT NOT NULL,
    User_Email VARCHAR(50) NOT NULL,
    User_Name CHAR(50) NOT NULL,
    User_Age INT NOT NULL,
    Password VARCHAR(50) NOT NULL,
    User_Occupation_ CHAR(50) NOT NULL,
    User_address VARCHAR(100) NOT NULL,
    PRIMARY KEY (user_ID)
);

CREATE TABLE categories_of_event
(
    ID INT NOT NULL,
    name_of_event CHAR(100) NOT NULL,
    Terms_and_codtions VARCHAR(500) NOT NULL,
    type_of_Event CHAR(100) NOT NULL,
    PRIMARY KEY (ID)
);

CREATE TABLE event
(
    event_name VARCHAR(100) NOT NULL,
    event_ID INT NOT NULL,
    event_place CHAR(100) NOT NULL,
    start_time INT NOT NULL,
    end_time INT NOT NULL,
    Event_date DATE NOT NULL,
    ID INT NOT NULL,
    PRIMARY KEY (event_ID),
    FOREIGN KEY (ID) REFERENCES categories_of_event(ID)
);

CREATE TABLE reminder
(
    ID INT NOT NULL,
    Description VARCHAR(300) NOT NULL,
    Status VARCHAR(100) NOT NULL,
    user_ID INT NOT NULL,
    event_ID INT NOT NULL,
    PRIMARY KEY (ID),
    FOREIGN KEY (user_ID) REFERENCES users(user_ID),
    FOREIGN KEY (event_ID) REFERENCES event(event_ID),
    UNIQUE (user_ID, event_ID)
);
```

THIS IS TABLE CODE

Data Insertion Examples:

```
--
--
-- INSERT INTO users(user_ID, User_Email,User_Name, User_Age, Password, User_Occupation_, User_address) VALUES
-- (1,'JHONSON123@gmail.com', 'JHONSON',20,'jhonson147', 'Student', '22,Gulsan');
-- INSERT INTO users(user_ID, User_Email,User_Name, User_Age, Password, User_Occupation_, User_address) VALUES
-- (2,'SON123@gmail.com', 'SON',20,'son17', 'Student', '20,Gulsan');
-- INSERT INTO users(user_ID, User_Email,User_Name, User_Age, Password, User_Occupation_, User_address) VALUES
-- (3,'JHON321@gmail.com', 'JHON',20,'son007', 'Student', '01,Danmodi');
-- INSERT INTO users(user_ID, User_Email,User_Name, User_Age, Password, User_Occupation_, User_address) VALUES
-- (4,'SMITH222@gmail.com', 'SMITH',20,'SMITH770', 'Student', '10,Danmodi');
-- INSERT INTO users(user_ID, User_Email,User_Name, User_Age, Password, User_Occupation_, User_address) VALUES
-- (5,'EVA555@gmail.com', 'EVA',20,'EVA779', 'Student', '10,Bannani');
-- select*from users
--
-- INSERT INTO categories_of_event(ID,name_of_event, Terms_and_codtions,type_of_Event) VALUES
-- (11,'night song', 'FOR EVERYONE AND NO PETS ALLOW', 'SONG');
-- INSERT INTO categories_of_event(ID,name_of_event, Terms_and_codtions,type_of_Event) VALUES
-- (22,'student galaday', 'FOR student of utb', 'General');
-- INSERT INTO categories_of_event(ID,name_of_event, Terms_and_codtions,type_of_Event) VALUES
-- (33,'SCIENCE FAIR', 'ONLY FOR of utb FAI', 'EDUCATION');
-- INSERT INTO categories_of_event(ID,name_of_event, Terms_and_codtions,type_of_Event) VALUES
-- (44,'MOVIE', 'FOR EVERYONE AND NO PETS ALLOW', 'INTERTENMENT');
-- select*from categories_of_event
--
-- INSERT INTO event(event_name,event_ID, event_place,start_time,end_time,Event_date,ID) VALUES
-- ('night song',10, 'ZLIN',9,11, '2025-06-25',11);
-- INSERT INTO event(event_name,event_ID, event_place,start_time,end_time,Event_date,ID) VALUES
-- ('student galaday',20, 'ZLIN',4,10, '2025-06-20',22);
-- INSERT INTO event(event_name,event_ID, event_place,start_time,end_time,Event_date,ID) VALUES
-- ('SCIENCE FAIR',30, 'ZLIN',9,11, '2025-06-15',33);
-- INSERT INTO event(event_name,event_ID, event_place,start_time,end_time,Event_date,ID) VALUES
-- ('MOVIE',40, 'ZLIN',9,11, '2025-05-15',44);
-- select*from event
--
-- INSERT INTO reminder(ID,Description, Status,user_ID,event_ID) VALUES
-- (11,'YOU HAVE AN EVENT TODAY', 'APPROVED',1,10);
-- INSERT INTO reminder(ID,Description, Status,user_ID,event_ID) VALUES
-- (22,'YOU HAVE AN EVENT WEEK', 'DISAPPROVED',2,20);
-- INSERT INTO reminder(ID,Description, Status,user_ID,event_ID) VALUES
-- (33,'YOU HAVE AN EVENT TODAY', 'APPROVED',3,30);
-- INSERT INTO reminder(ID,Description, Status,user_ID,event_ID) VALUES
-- (44,'YOU HAVE AN EVENT WEEK', 'APPROVED',4,40);
-- select*from reminder
```

THIS IS DATA ENTRY PART

1.

```
--(1) add columns to the USERS table.  
alter table users  
add Birthday date  
select*from users
```

146 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address	Birthday
1	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan	NULL
2	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan	NULL
3	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi	NULL
4	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi	NULL
5	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani	NULL

2.

```
--(2)DROP COLUMN data  
ALTER TABLE users  
drop COLUMN Birthday;  
  
select*from users
```

146 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address
1	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan
2	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan
3	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi
4	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi
5	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani

3.

```
--(3) UPDATE VALUE
UPDATE users
SET User_Age =22,
    User_Name = 'Sojib Hasan',
    User_Email = 's_hasan@utb.cz'
where user_ID = 1
select*from users
```

146 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address
1	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan
2	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan
3	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi
4	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi
5	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani

4.

```
--(4) WHERE
SELECT*FROM event
where end_time = 10;
```

146 %

Results Messages

	event_name	event_ID	event_place	start_time	end_time	Event_date	ID
1	student galaday	20	ZLIN	4	10	2025-06-20	22

5.

```
--(5) AND example
SELECT * FROM reminder
WHERE Description = 'APPROVED' AND ID = 11
```

146 %

Results Messages

ID	Description	Status	user_ID	event_ID
----	-------------	--------	---------	----------

6.

```
--(6) OR example
SELECT * FROM users
WHERE User_Age = 20 OR User_Occupation_ = 'Student';
```

146 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address
1	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan
2	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan
3	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi
4	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi
5	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani

7.

```
--(7) ORDER BY example
SELECT * FROM users
ORDER BY User_Name ASC;
```

146 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address
1	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani
2	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi
3	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi
4	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan
5	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan

8.

```
-- ONE MORE EXAMPLE
SELECT * FROM users
ORDER BY User_Age;
```

146 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address
1	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan
2	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi
3	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi
4	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani
5	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan

9.

```
--(9) example
select event_ID , event_place
from event
where start_time=9
```

146 %

Results Messages

	event_ID	event_place
1	10	ZLIN
2	30	ZLIN
3	40	ZLIN

10.

```
--(10) Add the country column (varchar(50)) to the user table

ALTER TABLE users
add country varchar(50);

select*from users
```

.46 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address	country
1	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan	NULL
2	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan	NULL
3	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi	NULL
4	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi	NULL
5	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani	NULL

11.

```
update users
set country = 'BANGLADESH'
where user_ID = 1

select*from users
```

146 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address	country
1	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan	BANGLADESH
2	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan	NULL
3	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi	NULL
4	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi	NULL
5	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani	NULL

B.

```
update users
set country = 'BANGLADESH'
where user_ID = 1

select*from users

update users
set country = 'BANGLADESH'
where user_ID = 3

update users
set country = 'czech republic'
where user_ID = 2

update users
set country = 'BANGLADESH'
where user_ID = 4

update users
set country = 'US'
where user_ID = 5

SELECT*FROM users
```

21 %

Results Messages

	user_ID	User_Email	User_Name	User_Age	Password	User_Occupation_	User_address	country
1	1	s_hasan@utb.cz	Sojib Hasan	22	jhonson147	Student	22,Gulsan	BANGLADESH
2	2	SON123@gmail.com	SON	20	son17	Student	20,Gulsan	czech republic
3	3	JHON321@gmail.com	JHON	20	son007	Student	01,Danmodi	BANGLADESH
4	4	SMITH222@gmail.com	SMITH	20	SMITH770	Student	10,Danmodi	BANGLADESH
5	5	EVA555@gmail.com	EVA	20	EVA779	Student	10,Bannani	US

12.

```
--12. List the USER name whose age is 20
```

```
select User_Name , user_ID, User_Email, User_address  
from users  
where User_Age like 20
```

146 %

Results Messages

	User_Name	user_ID	User_Email	User_address
1	SON	2	SON123@gmail.com	20,Gulsan
2	JHON	3	JHON321@gmail.com	01,Danmodi
3	SMITH	4	SMITH222@gmail.com	10,Danmodi
4	EVA	5	EVA555@gmail.com	10,Bannani

13.

```
--14 List all events happening in June 2025
```

```
SELECT event_name, event_place, Event_date  
FROM event  
WHERE Event_date BETWEEN '2025-06-01' AND '2025-06-30'  
ORDER BY Event_date;
```

146 %

Results Messages

	event_name	event_place	Event_date
1	SCIENCE FAIR	ZLIN	2025-06-15
2	student galaday	ZLIN	2025-06-20
3	night song	ZLIN	2025-06-25

14.

```
--14 List all events happening in June 2025
```

```
SELECT event_name, event_place, Event_date  
FROM event  
WHERE Event_date BETWEEN '2025-06-01' AND '2025-06-30'  
ORDER BY Event_date;
```

146 %

Results Messages

	event_name	event_place	Event_date
1	SCIENCE FAIR	ZLIN	2025-06-15
2	student galaday	ZLIN	2025-06-20
3	night song	ZLIN	2025-06-25

15.

```
-- 15 Count how many users are from each country
```

```
SELECT country, COUNT(*) AS Number_of_Users  
FROM users  
GROUP BY country  
ORDER BY Number_of_Users DESC;
```

146 %

Results Messages

	country	Number_of_Users
1	BANGLADESH	3
2	czech republic	1
3	US	1

MODIFY CODE

WORK WITH VIEWS

--User Event Summary

```
-- MODIFY CODE
-- WORK WITH VIEWS

CREATE VIEW UserEventSummary AS
SELECT u.User_Name, u.User_Email,
       COUNT(r.event_ID) AS Total_Events,
       SUM(CASE WHEN r.Status = 'APPROVED' THEN 1 ELSE 0 END) AS Approved_Events
FROM users u
LEFT JOIN reminder r ON u.user_ID = r.user_ID
GROUP BY u.User_Name, u.User_Email;

SELECT*FROM UserEventSummary
```

146 %

Results Messages

	User_Name	User_Email	Total_Events	Approved_Events
1	EVA	EVA555@gmail.com	0	0
2	JHON	JHON321@gmail.com	1	1
3	Sojb Hasan	s_hasan@utb.cz	1	1
4	SMITH	SMITH222@gmail.com	1	1
5	SON	SON123@gmail.com	1	0

WORK WITH TRIGGERS

--Update Last Modified Date

A.

```
-- MODIFY CODE
-- WORK WITH TRIGGERS
--Update Last Modified Date

--column add
ALTER TABLE event
ADD last_modified DATETIME DEFAULT GETDATE();

select*from event
```

146 %

Results Messages Client Statistics

	event_name	event_ID	event_place	start_time	end_time	Event_date	ID	last_modified
1	night song	10	ZLIN	9	11	2025-06-25	11	NULL
2	student galaday	20	ZLIN	4	10	2025-06-20	22	NULL
3	SCIENCE FAIR	30	ZLIN	9	11	2025-06-15	33	NULL
4	MOVIE	40	ZLIN	9	11	2025-05-15	44	NULL

```
-- MODIFY CODE
-- WORK WITH TRIGGERS
--Update Last Modified Date

--column add
ALTER TABLE event
ADD last_modified DATETIME DEFAULT GETDATE();

select*from event

-- new trigger

CREATE TRIGGER UpdateEventModifiedDate
ON event
AFTER UPDATE
AS
BEGIN
    UPDATE event
    SET last_modified = GETDATE()
    WHERE event_ID IN (SELECT event_ID FROM inserted)
END;

-- now it's test
-- like upadte a data in table

UPDATE event
SET event_place = 'New Location'
WHERE event_ID = 10;

-- is it true or not

SELECT event_ID, event_name, last_modified
FROM event
WHERE event_ID = 10;
```

91 %

Results Messages Client Statistics

	event_ID	event_name	last_modified
1	10	night song	2025-05-09 12:47:11.653

--STORED PROCEDURES
--Get User's Monthly Events

SQLQuery1.sql - dat...ojekt (A24672 (52))

```
--STORED PROCEDURES
--Get User's Monthly Events

CREATE PROCEDURE GetUserMonthlyEvents
    @UserID INT,
    @Month INT,
    @Year INT
AS
BEGIN
    SELECT e.event_name, e.Event_date, e.start_time, e.end_time,
           c.type_of_Event, r.Status
    FROM event e
    JOIN reminder r ON e.event_ID = r.event_ID
    JOIN categories_of_event c ON e.ID = c.ID
    WHERE r.user_ID = @UserID
           AND MONTH(e.Event_date) = @Month
           AND YEAR(e.Event_date) = @Year
    ORDER BY e.Event_date, e.start_time;
END;
SELECT*FROM reminder
EXEC GetUserMonthlyEvents 1, 6, 2025;
```

146 %

Results Messages

	event_name	Event_date	start_time	end_time	type_of_Event	Status
1	night song	2025-06-25	9	11	SONG	APPROVED

```
--FUNCTIONS
--Calculate Event Duration in Hours
```

SQLQuery1.sql - dat...ojekt (A24672 (52))

```
--FUNCTIONS
--Calculate Event Duration in Hours

CREATE FUNCTION CalculateEventDuration(@EventID INT)
RETURNS DECIMAL(5,2)
AS
BEGIN
    DECLARE @Duration DECIMAL(5,2);

    SELECT @Duration = (end_time - start_time) / 100.0
    FROM event
    WHERE event_ID = @EventID;

    RETURN @Duration;
END;

select*from event

SELECT dbo.CalculateEventDuration(10) AS 'Duration (Hours)';
```

146 %

Results Messages Client Statistics

	Duration (Hours)
1	0.02

CONCLUTION:

A strong and expandable basis for handling individual scheduling, event monitoring, and reminder notifications is established by the Personal Calendar database system that has been put into place. Through suitable primary and foreign key restrictions with little redundancy, the relational database design ensures data integrity while adhering to normalization rules and preserving optimal performance for individual users.

With a design that easily accommodates future AI-enhanced features like intelligent scheduling recommendations, the project effectively translates academic database theory into a useful personal productivity tool. Other improvements can consist of:

- *Full-text search for event content
- *End-to-end encryption for sensitive reminders
- *Natural language processing for voice input

FINISH